Scheduling Learned Image Codes over Fading Multi-User Downlinks: A Link Abstraction and a Calibration-Aware Restless-Bandit Scheduler

First Author and Second Author

***Abstract*—Deep joint source-channel coding (JSCC) is designed and evaluated on single links, yet a cellular downlink serves many users slot by slot, through a scheduler, before a deadline. Scheduling needs each slot's value to each user; no model provides it for a learned code. We show that a progressive learned code trained under slot-varying SNR obeys a link abstraction: its distortion is the source distortion plus a sensitivity-weighted sum of per-slot minimum mean-square-error terms. On two test sets it predicts held-out decodes within 0.055–0.083 dB, against 0.16–0.19 dB for EESM and 0.29–0.34 dB for MIESM given the same weights; a constant-SNR-trained code obeys no such rule. Calibrated from 12 decodes per image, the abstraction reduces each user to a two-number state, and deadline scheduling becomes a finite-horizon restless bandit. We propose CLASP, a calibration-aware, link-abstraction-based scheduler that solves it with exact per-user dynamic programs, a Lagrangian bound on every causal scheduler and an index policy, and rewards the probability of reaching a quality target under the measured error of its own prediction, from the abstraction and reports. Decoded, CLASP gains 0.5–0.6 dB of mean PSNR over proportional fair and brings an extra 8–15% of users to 30 dB over per-image rate selection, with fresh, noisy or aged reports. Against digital transmission with CQI-table link adaptation and HARQ it leads by 1.2–1.6 dB and 13–16% of users at 30 dB under 2 dB CQI errors or aged reports. Training the code on the channel CLASP creates adds 0.21 dB.**

***Index Terms*—Deep joint source-channel coding, hybrid automatic repeat request, image transmission, imperfect channel state information, index policies, Lagrangian relaxation, link abstraction, multi-user scheduling, restless bandits.**

I. Introduction

DEEP joint source-channel coding (JSCC) maps an image straight to channel symbols with one learned encoder and decoder, with no separate source and channel codes [1]. It matches or beats separation-based digital transmission at low SNR and degrades gracefully as the channel worsens, instead of failing once the SNR drops below what a digital code was designed for [1], [2]. That makes it a strong candidate for image delivery in 6G [2]. But a learned code is designed and evaluated as a single link at a given SNR: one transmitter, one receiver, the channel condition handed to it [1], [3], [4]. A cellular downlink is different. A base station splits a frame of slots among many users through a scheduler; each user's slots fade independently; the base station knows only an estimate of the current slot, a channel-quality report [5] that is often late or wrong; and each delivery, one image, has a deadline. So the gain a point-to-point study reports is not automatically the gain a cell delivers. That depends on the scheduler, and no existing scheduler knows what a slot is worth to a learned code. To schedule, the base station needs the value of "give user k this slot now." For a digital code that value follows from the rate the slot carries and the codec's rate-distortion curve. For a trained, nonlinear code no such value exists.

Existing approaches to serving images to many users over a fading downlink fall into two categories (Fig. 1): *physical link design*, which keeps the image codec separate and engineers the link that carries its bits, and *the JSCC way*, which lets a learned code take on the channel itself.

Physical link design is how cellular systems deliver images today. A standard codec compresses the image into bits once (HEVC intra, as in HEIC [6]); for each slot, the physical layer picks the modulation and coding rate from the user's channel quality indicator (CQI) report [5], [7]; failed blocks are retransmitted with incremental-redundancy hybrid automatic repeat request (HARQ) [8]; and a channel-aware scheduler, such as proportional fair (PF), picks the user each slot serves [9]. Its tools for many users are mature: effective-SNR link abstractions, the exponential and the mutual-information effective SNR mappings (EESM, MIESM), predict a coded block's error rate from the SNRs of all the resources it occupies [10]–[12], and media-aware schedulers weigh each user's next packet by the distortion it removes before its deadline [13]–[16]. All of them, however, rest on a binary outcome per block, decoded or lost, so a user's quality is a step function of its channel: a slot below the block's design SNR delivers nothing, and SNR above it is wasted, the cliff effect [1], [2]. Moreover, the rate is committed before the slot is seen, from a report that may be late or wrong, and link adaptation and HARQ protect the bits only by spending extra slots. In our simulations, going from fresh reports to aged ones (correlation 0.8) costs the digital chain 1.9 dB of mean PSNR, against 0.6 dB for a learned code under our scheduler (Section V-D).

In contrast, the JSCC way replaces the two codes with one learned code that maps the image straight to channel symbols [1]. Its quality degrades smoothly with the SNR its symbols meet, so no block is lost outright and no coding rate has to be matched to an uncertain report; the same symbols serve receivers of different SNRs, as linear analog codes showed for multicast video [17]. One line of work adapts a single link to the SNR [3], to the bandwidth and rate [4], [18]–[20], to feedback that stops the transmission once a target quality is reached [21], [22], and to MIMO channels [23]. To serve many users, the JSCC way splits the shared channel

Manuscript created November 2026.

First Author and Second Author are with [department], [university], [city], [country] (e-mail: [address]).

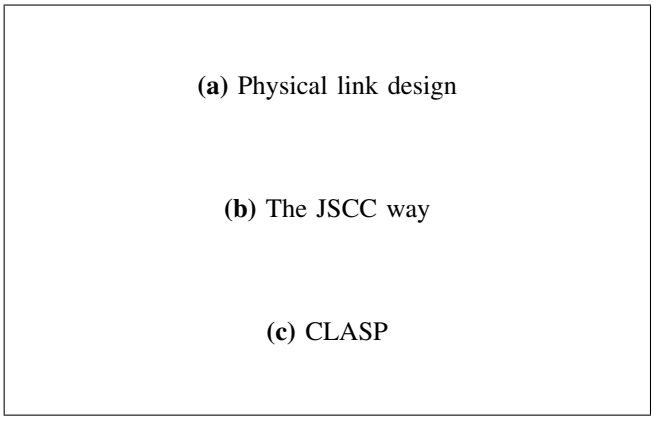


Fig. 1. Approaches to serving images to many users over a fading downlink. (a) Physical link design: a separate codec over adaptive modulation and coding, HARQ and a channel-aware scheduler; each block is decoded or lost. (b) The JSCC way: a learned code degrades gracefully, but each user is valued at a single SNR and allocated once, or scheduled by a policy learned by trial and error. (c) CLASP: a progressive learned code trained under slot-varying SNR obeys a link abstraction that values every slot, and a restless-bandit scheduler built on it serves every user slot by slot up to the deadline.

by each user's predicted quality at its SNR, through channel and symbol assignment [24], user association and bandwidth [25], or Bayesian optimization over a learned quality oracle [26]; other work builds the sharing into the code [27]. Where the allocation must change over time, a scheduler is learned by deep reinforcement learning [28], [29], most recently as a restless bandit over fading slots whose slot values are learned [30]. Our method belongs to the JSCC way: it schedules a learned code to many users across fading slots.

Although existing JSCC methods are effective, they face three challenges when a base station schedules a learned code to many users over fading slots.

- *No link abstraction.* They value a user's transmission at a single SNR, with a fitted quality-SNR curve or a learned quality predictor [20], [24], [26]. A scheduled image, however, is spread over slots of different SNRs, and for a nonlinear learned code no rule is known for how these SNRs combine into decoded quality. EESM and MIESM predict a digital block's error rate, not a learned code's distortion, and to our knowledge have not been validated on a learned code.
- *No tractable, certifiable scheduler.* Deciding which user each slot serves, before a deadline and without knowing future slots, is a stochastic dynamic program whose state grows exponentially with the number of users. Existing methods either decide each transmission at its current SNR without planning over the slots left before the deadline [24]–[26], or learn slot values by reinforcement learning, which needs many interactions to train and gives no certificate of how far the learned policy is from optimal [28]–[30].
- *Imperfect information.* The quality model and the channel reports are both estimates, yet existing schedulers take them at face value: per-image rate selection (PADC [20]) treats its predicted quality as exact, and the learned schedulers assume perfect channel knowledge [30], so a user predicted just above a quality target can decode just below it. Moreover, the advantage of the JSCC way over physical link design was measured on single links against an idealized digital side, with exact channel knowledge and capacity-achieving codes [21]; whether it survives the reports a real base station has is open.

Behind all three lies one question: does a learned code have a link abstraction at all, a rule that, like EESM for a digital block, turns the SNRs its symbols met into its decoded quality? And if it does, what decides whether a code obeys it?

To answer it, we trained a progressive code, which decodes from any prefix of its phases, under slot-varying SNR, and tested candidate rules on held-out decodes with mixed slot SNRs (Sections IV-B and V-A). We found three things. First, its distortion adds up over slots: the source distortion plus a sensitivity-weighted sum of per-slot minimum mean-square-error (MMSE) terms predicts held-out decodes with a mean absolute error of 0.055–0.083 dB on two test sets, against 0.16–0.19 dB for EESM and 0.29–0.34 dB for MIESM with the same per-phase weights. Linear analog codes obey this form by construction [17]; here a nonlinear learned code acquires it. Second, training decides it: the same architecture trained at one constant SNR per image obeys no such rule (the MMSE rule errs by 0.22 dB, and no rule does better than 0.15 dB), because it codes its later phases against the base reconstruction it expects at one SNR, the mismatch DeepJSCC-l observed for layers [18]. Third, the abstraction is cheap and accurate, yet its small error matters. Twelve decodes per image calibrate it, and it predicts the decoded quality of complete scheduled deliveries with a mean absolute error of 0.06–0.09 dB and a bias within ± 0.03 dB, with exact, noisy or aged reports. At a hard quality target, however, a scheduler that takes the calibrated curves as exact leaves an extra 1.7–2.0% of the users below 30 dB, at every cell size from 16 to 128 users. In short, training gives a learned code a link abstraction. With it, each user's state reduces to two numbers, the phases received and their accumulated weighted noise, and scheduling becomes a planning problem, provided the abstraction's residual error is part of the objective.

To address these challenges, we propose CLASP (Calibration-aware, Link-Abstraction-based Scheduling of Progressive codes), a scheduler for learned progressive codes in a multi-user fading downlink. Unlike existing JSCC methods, which value each user at one SNR or learn slot values by reinforcement learning, CLASP values every slot for every user through the code's link abstraction and plans each user's delivery to the deadline exactly. It has one design for each challenge.

- *Link abstraction.* The progressive code is trained under slot-varying SNR and its MMSE link abstraction is calibrated per image, so that each user becomes a two-number state.
- *Scheduler.* On this state the frame becomes a finite-horizon restless bandit [31]. A Lagrangian relaxation splits it into one small dynamic program per user under per-slot prices, solved exactly on a CPU; the relaxation's value is an upper bound on every causal scheduler; and users are served in the order of the Lagrangian index

those programs define [32], [33].

- *Imperfect information.* Each program rewards the probability of reaching the quality target under a calibrated predictive distribution of the user's decoded quality, rather than a hard threshold on the predicted quality. The distribution's spread is the measured error of the prediction, from the calibrated curves and from the channel reports. Given an aged report, each program decides on the posterior of the channel, not on the report itself.

The main contributions of this paper are as follows.

- *Discovery.* We find that a learned progressive code trained under slot-varying SNR obeys an additive MMSE link abstraction, more accurate than EESM and MIESM, while a code trained at constant SNR obeys none, and that a transmitter can calibrate it from 12 decodes per image. To our knowledge, effective-SNR mappings had not been tested on a learned code before.
- *Method.* We propose CLASP, which casts multi-user deadline scheduling of learned codes as a finite-horizon restless bandit with a two-number user state, solves it with exact per-user programs, a Lagrangian bound and an index policy, and is robust to the errors of its own quality prediction and of the channel reports.
- *Performance.* Decoded by the codec, CLASP gains 0.5–0.6 dB of mean PSNR over PF and brings an extra 8–15% of the users to 30 dB over per-image rate selection, with fresh, noisy or aged reports. In simulation its gap to the bound vanishes as the cell grows: from 0.21 to 0.005 dB per user between 8 and 128 users for mean PSNR, and for the target within $\pm 0.3\%$ of the bound from 16 users on. Against a digital chain with CQI-table link adaptation and HARQ, it leads at every CSI condition tested, by 1.2–1.6 dB and an extra 13–16% of the users at 30 dB with 2 dB CQI errors or reports aged to correlation 0.8. An idealized digital code 3 dB from capacity with the same HARQ leads CLASP by 2.4% of the users with exact reports and trails it by 5–14% from 1 dB CQI errors or correlation 0.95 on. Training the code on the channel CLASP creates adds 0.21 dB, decoded.

The rest of the paper is organized as follows. Section II reviews related work, and Section III states the system model and the problem. Section IV presents the training of the progressive code, its MMSE link abstraction, CLASP, and the training of the code on the channel CLASP creates. Section V evaluates the abstraction, and CLASP against PF, per-image rate selection, the Lagrangian bound and the digital chain. Section VI concludes and states the limitations.

II. Related Work

A. Physical Link Design for Multi-User Media Delivery

B. The JSCC Way

C. Restless Bandits and Lagrangian Index Policies

III. Preliminaries: System Model and Problem Statement

IV. Method

A. Training the Progressive Code for Slot-Varying SNR

B. The MMSE Link Abstraction

C. CLASP: Deadline Scheduling as a Finite-Horizon Restless Bandit

D. Imperfect Information: The Calibration-Aware Reward and Aged Reports

E. Training the Code on the Channel CLASP Creates

V. Experiments

A. Link Abstraction Accuracy

B. Mean PSNR: Near the Bound, and over PF

C. Quality Target and Latency

D. Imperfect CSI and the Digital Baseline

E. Training the Code on CLASP's Channel

F. Ablations and System Choice

VI. Conclusion

References

[1] E. Bourtsoulatze, D. B. Kurka, and D. Gündüz, "Deep joint source-channel coding for wireless image transmission," *IEEE Trans. Cogn. Commun. Netw.*, vol. 5, no. 3, pp. 567–579, Sep. 2019.

[2] D. Gündüz, Z. Qin, I. E. Aguerri, H. S. Dhillon, Z. Yang, A. Yener, K. K. Wong, and C.-B. Chae, "Beyond transmitting bits: Context, semantics, and task-oriented communications," *IEEE J. Sel. Areas Commun.*, vol. 41, no. 1, pp. 5–41, Jan. 2023.

[3] J. Xu, B. Ai, W. Chen, A. Yang, P. Sun, and M. Rodrigues, "Wireless image transmission using deep source channel coding with attention modules," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 4, pp. 2315–2328, Apr. 2022.

[4] K. Yang, S. Wang, J. Dai, X. Qin, K. Niu, and P. Zhang, "SwinJSCC: Taming Swin transformer for deep joint source-channel coding," *IEEE Trans. Cogn. Commun. Netw.*, vol. 11, no. 1, pp. 90–104, Feb. 2025.

[5] 3GPP, "Evolved Universal Terrestrial Radio Access (E-UTRA); Physical layer procedures (Release 8)," 3rd Generation Partnership Project, TS 36.213 V8.8.0, Sep. 2009.

[6] G. J. Sullivan, J.-R. Ohm, W.-J. Han, and T. Wiegand, "Overview of the High Efficiency Video Coding (HEVC) standard," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 22, no. 12, pp. 1649–1668, Dec. 2012.

[7] A. J. Goldsmith and S.-G. Chua, "Variable-rate variable-power MQAM for fading channels," *IEEE Trans. Commun.*, vol. 45, no. 10, pp. 1218–1230, Oct. 1997.

[8] G. Caire and D. Tuninetti, "The throughput of hybrid-ARQ protocols for the Gaussian collision channel," *IEEE Trans. Inf. Theory*, vol. 47, no. 5, pp. 1971–1988, Jul. 2001.

[9] P. Viswanath, D. N. C. Tse, and R. Laroia, "Opportunistic beamforming using dumb antennas," *IEEE Trans. Inf. Theory*, vol. 48, no. 6, pp. 1277–1294, Jun. 2002.

[10] Ericsson, "System-level evaluation of OFDM – further considerations," 3GPP TSG-RAN WG1 #35, Lisbon, Portugal, Tdoc R1-031303, Nov. 2003.

[11] IEEE 802.16 Broadband Wireless Access Working Group, "IEEE 802.16m evaluation methodology document (EMD)," IEEE 802.16m-08/004r5, Jan. 2009.

[12] S. Lagen, K. Wanuga, H. Elkotby, S. Goyal, N. Patriciello, and L. Giupponi, "New Radio physical layer abstraction for system-level simulations of 5G networks," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Dublin, Ireland, Jun. 2020, pp. 1–7.

[13] P. A. Chou and Z. Miao, "Rate-distortion optimized streaming of packetized media," *IEEE Trans. Multimedia*, vol. 8, no. 2, pp. 390–404, Apr. 2006.
[14] P. Pahalawatta, R. Berry, T. Pappas, and A. Katsaggelos, "Content-aware resource allocation and packet scheduling for video transmission over wireless networks," *IEEE J. Sel. Areas Commun.*, vol. 25, no. 4, pp. 749–759, May 2007.
[15] F. Fu and M. van der Schaar, "A systematic framework for dynamically optimizing multi-user wireless video transmission," *IEEE J. Sel. Areas Commun.*, vol. 28, no. 3, pp. 308–320, Apr. 2010.
[16] S. A. Hosseini and S. S. Panwar, "Restless video bandits: Optimal SVC streaming in a multi-user wireless network," *IEEE Access*, vol. 7, pp. 150 805–150 822, 2019.
[17] S. Jakubczak and D. Katabi, "A cross-layer design for scalable mobile video," in *Proc. 17th Annu. Int. Conf. Mobile Comput. Netw. (MobiCom)*, Las Vegas, NV, USA, Sep. 2011, pp. 289–300.
[18] D. B. Kurka and D. Gündüz, "Bandwidth-agile image transmission with deep joint source-channel coding," *IEEE Trans. Wireless Commun.*, vol. 20, no. 12, pp. 8081–8095, Dec. 2021.
[19] J. Dai, S. Wang, K. Tan, Z. Si, X. Qin, K. Niu, and P. Zhang, "Nonlinear transform source-channel coding for semantic communications," *IEEE J. Sel. Areas Commun.*, vol. 40, no. 8, pp. 2300–2316, Aug. 2022.
[20] W. Zhang, H. Zhang, H. Ma, H. Shao, N. Wang, and V. C. M. Leung, "Predictive and adaptive deep coding for wireless image transmission in semantic communication," *IEEE Trans. Wireless Commun.*, vol. 22, no. 8, pp. 5486–5501, Aug. 2023.
[21] D. B. Kurka and D. Gündüz, "DeepJSCC-f: Deep joint source-channel coding of images with feedback," *IEEE J. Sel. Areas Inf. Theory*, vol. 1, no. 1, pp. 178–193, May 2020.
[22] H. Wu, Y. Shao, E. Ozfatura, K. Mikolajczyk, and D. Gündüz, "Transformer-aided wireless image transmission with channel feedback," *IEEE Trans. Wireless Commun.*, vol. 23, no. 9, pp. 11 904–11 919, Sep. 2024.
[23] H. Wu, Y. Shao, C. Bian, K. Mikolajczyk, and D. Gündüz, "Deep joint source-channel coding for adaptive image transmission over MIMO channels," *IEEE Trans. Wireless Commun.*, vol. 23, no. 10, pp. 15 002–15 017, Oct. 2024.
[24] L. Yan, Z. Qin, R. Zhang, Y. Li, and G. Y. Li, "Resource allocation for text semantic communications," *IEEE Wireless Commun. Lett.*, vol. 11, no. 7, pp. 1394–1398, Jul. 2022.
[25] L. Xia, Y. Sun, D. Niyato, X. Li, and M. A. Imran, "Joint user association and bandwidth allocation in semantic communication networks," *IEEE Trans. Veh. Technol.*, vol. 73, no. 2, pp. 2699–2711, Feb. 2024.
[26] H. Hou, S. Bi, X. Li, H. Zhang, and Z. Quan, "Reliable online resource allocation for multi-user semantic communications: A constraint Bayesian optimization approach," *IEEE Trans. Wireless Commun.*, vol. 25, pp. 22 323–22 338, 2026.
[27] S. F. Yilmaz, C. Karamanlı, and D. Gündüz, "Distributed deep joint source-channel coding over a multiple access channel," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Rome, Italy, May 2023, pp. 1400–1405.
[28] Y. Wang, M. Chen, T. Luo, W. Saad, D. Niyato, H. V. Poor, and S. Cui, "Performance optimization for semantic communications: An attention-based reinforcement learning approach," *IEEE J. Sel. Areas Commun.*, vol. 40, no. 9, pp. 2598–2613, Sep. 2022.
[29] H. Zhang, H. Wang, Y. Li, K. Long, and A. Nallanathan, "DRL-driven dynamic resource allocation for task-oriented semantic communication," *IEEE Trans. Commun.*, vol. 71, no. 7, pp. 3992–4004, Jul. 2023.
[30] C. McDowell, A. Panebianco, S. Chakraborty, and Y. Sun, "Significance-driven semantic communication," arXiv:2608.28441, Aug. 2026.
[31] P. Whittle, "Restless bandits: Activity allocation in a changing world," *J. Appl. Probab.*, vol. 25, no. A, pp. 287–298, 1988.
[32] L. Hao, Y. Xu, and L. Tong, "Asymptotically optimal Lagrangian priority policy for deadline scheduling with processing rate limits," *IEEE Trans. Autom. Control*, vol. 67, no. 1, pp. 236–250, Jan. 2022.
[33] D. B. Brown and J. E. Smith, "Index policies and performance bounds for dynamic selection problems," *Manage. Sci.*, vol. 66, no. 7, pp. 3029–3050, Jul. 2020.